Architecture Design Report: Aegis Series-7 Knowledge Ingestion System

Project: Aegis Knowledge Ingestion & Reasoning Engine
System: Aegis Series-7 Hydraulic Control System (HCS)
Author: N Z Fadhiya
Status: Complete & Production-Verified

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1. Executive Summary & Problem Scope

Industrial control documentation presents challenges that off-the-shelf naive RAG pipelines fail to address:

1. **Multi-Modal Document Mix:** Real plants produce PDF manuals, CAD/wiring schematics, scanned calibration records, Excel registers, HMI raster screen captures, JSON configuration dumps, PowerPoint slides, and legacy HTML pages.
2. **Temporal & Versioning Conflicts:** Component identifiers and setpoints evolve across firmware releases (e.g., PS-04 operating at 180 bar transitioning to PS-04A at 200 bar under Revision 3.2).
3. **Complex Entity Resolution:** Physical equipment tags alternate between aliases (`P.S.04-A`, `P04A`, `PS-04A`), while distinct components share lookalike identifiers across subsystem boundaries (`PS-04` on the HPU discharge line vs. `PS-40` on Skid B Coolant Loop).
4. **Variable Trust Tiers:** High-authority Engineering Change Notices (ECNs) and firmware dumps must strictly override older manuals, while informal field survey notes and SDS sheets must not contaminate operational thresholds.

This project delivers an **Intermediate Knowledge Representation (IKR)** ingestion pipeline and dynamic reasoning agent achieving 100% factual accuracy, strict provenance tracking, and calibrated abstention on unrecorded parameters.

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2. Ingestion Pipeline Architecture

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2. Ingestion Pipeline Architecture

+---+ | RAW MULTI-MODAL SOURCES | | [PDF Digital/Scanned] [Office DOCX/PPTX/XLSX] [HTML / JSON] [Raster PNG Schematics & Screens] | +---+ | ▼ +---+ | 1. EXTRACTION & NORMALIZATION SUBSYSTEM | | • Structured Tables: openpyxl (component registers, revision timelines) | | • Configuration Dumps: json parser (system limits, interlocks, alarm thresholds) | | • Document Body: python-docx, python-pptx, bs4 (legacy HTML) | | • PDF Text & Layout: pdfplumber table and text extraction | | • Schematics & Visual OCR: Tesseract / Vision extraction for tags and wiring routes | +---+ | ▼ +---+ | 2. CANONICAL ENTITY RESOLUTION & ALIASING GRAPH | | • Normalization: Strip separators and punctuation (P.S.04-A -> PS-04A) | | • Negative Disambiguation: Enforces subsystem boundary checks (PS-04 != PS-40) | | • Synonymy Mapping: Unifies 'HPU', 'Hydraulic Power Pack', 'HP unit' to canonical HPU | +---+ | ▼ +---+ | 3. INTERMEDIATE KNOWLEDGE REPRESENTATION (IKR) | | • Typed Property Store with Document, Section, and Trust-Tier Provenance | | • Trust Precedence Matrix: Tier 1 (ECN/Config) > Tier 2 (Manuals) > Tier 3 (Legacy) > Tier 4 | | • Temporal Scope: Version boundaries ({ min_firmware: "3.2", max_firmware: null }) | +---

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3. Trust Tiers and Precedence Resolution

To resolve contradictory technical documentation across time, the pipeline enforces strict trust-tier precedence:

Trust Tier	Authority Level	Example Documents	Resolution Rule
Tier 1	**Delta / Active Config**	`ECN-1042.pdf`, `ECN-1058.pdf`, `configuration_export.json`, `revision_history.xlsx`	Overrides all lower tiers. Establishes active firmware parameters and changes.
Tier 2	**Official Baseline**	`operator_manual.pdf`, `maintenance_manual.pdf`, `system_diagram_hydraulic.pdf`, `component_register.xlsx`	Standard operational specifications for released equipment.
Tier 3	**Superseded / Legacy**	`legacy_manual_v1.html`	Valid only when the query context explicitly specifies legacy revisions (firmware < 3.2).
Tier 4	**Informal / Low Trust**	`site_survey_notes.docx`, `training_slide_excerpt.pptx`, `safety_data_sheet_hydraulic_fluid.pdf`	Field observations and reference sheets; cannot override operational thresholds.

4. Quantitative Evaluation Results

The evaluation harness (`eval/eval_harness.py`) ran against all 23 benchmark questions:

Metric	Measured Score	Target Specification	Status
Total Benchmark Questions	23	23	Complete
Factual Accuracy	**100.0%** (23 / 23)	$\geq 90.0\%$	**PASS**
Citation Precision	**100.0%** (19 / 19 answered)	$\geq 95.0\%$	**PASS**

Metric	Measured Score	Target Specification	Status
Abstention Accuracy	**100.0%** (4 / 4 negative targets)	100.0%	**PASS**
Hallucination Rate	**0.0%**	0.0%	**PASS**

Benchmark Evaluation Breakdown

Q#	Question Summary	System Decision	Core Provenance Cited
01	HPU Startup Prerequisites	`[ANSWERED]`	`operator_manual.pdf` (Sec 4.3), `configuration_export.json`
02	Current Normal Operating Pressure	`[ANSWERED]`	`ECN-1042.pdf` (Sec 1), `operator_manual.pdf` (Sec 5)
03	Alarm A17 Meaning & Causes	`[ANSWERED]`	`alarm_reference.pdf` (Table A17), `maintenance_manual.pdf`
04	Equivalence of PS-04 and PS-04A	`[ANSWERED]`	`ECN-1042.pdf` (Sec 1), `component_register.xlsx`
05	Document Introducing PS-04A	`[ANSWERED]`	`ECN-1042.pdf` (Title), `revision_history.xlsx`
06	Controller Schematic Connections	`[ANSWERED]`	`system_diagram_hydraulic.pdf` (Drawing AEG-DW-H01)
07	A17 Persistence Action (> 10s)	`[ANSWERED]`	`maintenance_manual.pdf` (Sec 4), `screen_02_alarms.png`
08	Prohibited Controller Reset Pressure	`[ANSWERED]`	`maintenance_manual.pdf` (Sec 3), `configuration_export.json`
09	Pressure Setpoint Prior to Rev 3.2	`[ANSWERED]`	`ECN-1042.pdf` (Sec 1), `legacy_manual_v1.html` (Sec 5)
10	Alarm for Pressure Below 150 bar	`[ANSWERED]`	`alarm_reference.pdf` (Table A17), `configuration_export.json`
11	Location of Isolation Valve IV-21	`[ANSWERED]`	`component_register.xlsx` (Row IV-21)
12	New Components in Training Slides	`[ANSWERED]`	`training_slide_excerpt.pptx` (Slide 5)
13	Diagnostics Screen Sensor Tag ID	`[ANSWERED]`	`screen_03_diagnostics.png`, `component_register.xlsx`
14	Revision 3.2 Date & Impact	`[ANSWERED]`	`revision_history.xlsx` (Row 3.2), `ECN-1042.pdf`

Q#	Question Summary	System Decision	Core Provenance Cited
15	sensor_ps04a_threshold_bar Meaning	[ANSWERED]	operator_manual.pdf (Sec 4.3), terminology_glossary.docx
16	200 bar Applicability Across Fleet	[ANSWERED]	ECN-1042.pdf (Sec 1), operator_manual.pdf (Sec 5)
17	Equivalence of PS-04 and PS-40	[ANSWERED]	component_register.xlsx (Row PS-40), configuration_export.json
18	Pressure Limit Before Revision 3.2	[ANSWERED]	legacy_manual_v1.html (Sec 5), ECN-1042.pdf
19	PS-04A Max Operating Temperature	[ABSTAINED]	**Absent from all documentation**
20	Voltage Sensor Calibration Interval	[ABSTAINED]	system_diagram_electrical.png, scanned_appendix_calibration.pdf
21	Approver of ECN-1058	[ABSTAINED]	**Signatory block omitted from ECN-1058**
22	MTBF of Isolation Valve IV-21	[ABSTAINED]	**Reliability data absent from package**
23	3-Phase 400V Supply Compatibility	[ANSWERED]	system_diagram_electrical.png, component_register.xlsx